



Stomach & Small Intestine – MCQ Practice

Anatomy GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A CT scan is reviewed to locate the pyloric opening of the stomach. At which vertebral level does it normally lie?

EASY

- A. T10
- B. T11
- C. L1 (transpyloric plane)
- D. L3
- E. T12

2. Which part of the duodenum is the only part supplied by end arteries and the only part devoid of circular mucosal folds?

EASY

- A. First (superior) part
- B. Second (descending) part
- C. Third (horizontal) part
- D. Fourth (ascending) part
- E. Duodenojejunal flexure

3. During a laparotomy, a surgeon searches for a Meckel's diverticulum. Where should he look?

EASY

- A. 2 cm proximal to the duodenojejunal flexure
- B. On the medial wall of the second part of the duodenum
- C. About 60 cm distal to the duodenojejunal flexure
- D. At the ileocecal junction itself
- E. About 60 cm (2 feet) proximal to the ileocecal junction

4. The greater omentum can seal off a site of infection. What are the dense aggregations of macrophages within it called?

EASY

- A. Peyer's patches
- B. Milky spots
- C. Brunner's glands
- D. White windows
- E. Arterial arcades

5. A 50-year-old man with years of epigastric pain presents with haematemesis and hypotension. Endoscopy shows a deep ulcer on the posterior wall of the first part of the duodenum with a bleeding vessel in its base. Which artery has most likely been eroded?

MEDIUM

- A. Left gastric artery
- B. Splenic artery
- C. Right gastroepiploic artery
- D. Gastroduodenal artery
- E. Superior mesenteric artery

6. A 55-year-old woman with a known gastric ulcer develops constant pain referred to her back, followed by massive haematemesis. The ulcer is on the posterior wall and has eroded into the pancreas. Which artery is most likely bleeding?

MEDIUM

- A. Gastroduodenal artery
- B. Splenic artery
- C. Left gastric artery
- D. Right gastric artery
- E. Inferior pancreaticoduodenal artery

7. A 64-year-old man has weight loss and early satiety. On examination there is a hard, enlarged lymph node just above the left clavicle. By which route did tumour cells most likely reach this node?

MEDIUM

- A. Thoracic duct
- B. Left gastric vein to the portal vein
- C. Right lymphatic duct
- D. Hemiazygos vein
- E. Direct spread through the diaphragm

8. A surgeon dissects the free margin of the lesser omentum (hepatoduodenal ligament). Which structure lies deepest (most posterior) in this margin?

MEDIUM

- A. Bile duct
- B. Hepatic artery proper
- C. Right gastric artery
- D. Left gastric vein
- E. Portal vein

9. A patient with gastritis feels the pain in the epigastric region of the abdominal wall. The stomach's sympathetic supply comes from which spinal segments, explaining this referral?

MEDIUM

- A. T1-T4
- B. T10-L1
- C. T6-T9
- D. L1-L2
- E. S2-S4

10. During a truncal vagotomy for peptic ulcer disease, the surgeon identifies the anterior vagal trunk as it enters the abdomen. Which statement about this trunk is correct?

MEDIUM

- A. It is formed by the right vagus and supplies the posterior surface
- B. It carries sympathetic fibres from the celiac plexus
- C. It is formed equally by both vagi and supplies only the pylorus
- D. It is formed by the left vagus and supplies the anterior surface of the stomach
- E. It is formed by the left vagus and supplies the posterior surface

11. At laparotomy, a surgeon examines a loop of small bowel. It is wide, thick-walled and deep red; its mesentery shows few arcades, long vasa recta and translucent 'windows' near the gut wall. Which segment is this?

MEDIUM

- A. Jejunum
- B. Ileum
- C. Terminal ileum
- D. Third part of the duodenum
- E. Transverse colon

12. During ERCP, the endoscopist sees a small opening about 2 cm superior to the major duodenal papilla on the posteromedial wall of the second part of the duodenum. What does it most likely drain?

MEDIUM

- A. Common bile duct
- B. Main pancreatic duct
- C. Cystic duct
- D. Hepatopancreatic ampulla
- E. Accessory pancreatic duct (of Santorini)

13. An endoscopy report places a bleeding source proximal to the ligament of Treitz. Which statement about this ligament is correct?

MEDIUM

- A. It suspends the pylorus from the left crus of the diaphragm
- B. It suspends the duodenojejunal flexure from the right crus of the diaphragm and marks the upper/lower GI boundary
- C. It anchors the ileocecal junction to the right sacroiliac joint
- D. It forms the free margin of the lesser omentum
- E. It is a peritoneal fold that carries the superior mesenteric artery

14. A CT shows a mass in the stomach bed pushing the stomach forward. All of the following form the stomach bed EXCEPT:

MEDIUM

- A. Spleen
- B. Left suprarenal gland
- C. Left lobe of the liver
- D. Body of the pancreas
- E. Transverse mesocolon

15. A 19-year-old woman who lost weight rapidly after a prolonged illness presents with post-prandial epigastric pain, nausea and vomiting. Imaging shows a dilated proximal duodenum with an abrupt cut-off in the midline at L3. What is the most likely mechanism?

HARD

- A. Kinking of the duodenojejunal flexure by the ligament of Treitz
- B. Compression of the second part by the gallbladder
- C. Obstruction at the major duodenal papilla
- D. Compression of the fourth part against the left psoas major
- E. Compression of the third part between the superior mesenteric artery and the abdominal aorta

16. A 30-year-old man has sudden epigastric pain that later localises to the right iliac fossa. Appendicitis is suspected, but surgery reveals a perforated ulcer on the anterior wall of the first inch of the duodenum. How did the fluid reach the right iliac fossa?

HARD

- A. It entered the greater sac above the transverse colon and ran down the right lateral paracolic gutter
- B. It entered the lesser sac and left through the epiploic foramen into the pelvis
- C. It tracked retroperitoneally along the right psoas major
- D. It passed along the root of the mesentery to the left iliac fossa
- E. It tracked inside the lesser omentum to the ileocecal junction

17. A driver suffers blunt upper-abdominal trauma from a steering wheel. Which duodenal part is most vulnerable to external injury, and which is best protected?

HARD

- A. Most vulnerable: first part; best protected: fourth part
- B. Most vulnerable: second part; best protected: third part
- C. Most vulnerable: fourth part; best protected: first part
- D. Most vulnerable: third part; best protected: second part (in the paravertebral gutter)
- E. All parts are equally vulnerable because the duodenum is retroperitoneal

18. A tumour arises in the wall of the third part of the duodenum. Which artery mainly supplies this region, and to which node group does its lymph ultimately drain?

HARD

- A. Superior pancreaticoduodenal artery; celiac nodes
- B. Inferior pancreaticoduodenal artery (from the SMA); superior mesenteric nodes
- C. Gastroduodenal artery; gastroduodenal then celiac nodes
- D. Right gastroepiploic artery; right gastroepiploic nodes
- E. Jejunal arteries; mesenteric nodes only

19. A surgeon traces the root of the small-intestine mesentery from the duodenojejunal flexure (left side of L2) to the right sacroiliac joint. Which structure is NOT crossed by the root?

HARD

- A. Third part of the duodenum
- B. Inferior vena cava
- C. Left ureter
- D. Right gonadal vessels
- E. Right psoas major

20. During removal of the pyloric antrum, the surgeon incises the greater omentum parallel and inferior to an artery on the greater curvature and ligates all its omental branches. The remaining stomach still survives. Which artery is this, and why does the remnant survive?

HARD

- A. Left gastric artery; it arises directly from the celiac trunk
- B. Short gastric arteries; they come from the splenic artery
- C. Left gastroepiploic artery; the stomach has no arterial anastomoses
- D. Right gastroepiploic (gastro-omental) artery; anastomoses between the gastric arteries provide collateral supply
- E. Right gastric artery; its branches are end arteries

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	The pyloric opening lies at L1 on the transpyloric plane, about 1.5 cm right of the median plane; the cardiac opening is at T11.
2	A	EASY	The first part is unique on both counts, which is part of why it is the commonest site of duodenal ulcers.
3	E	EASY	Rule of twos: 2% of people, 2 inches long, 2 feet (60 cm) proximal to the ileocecal junction. Medial wall of D2 is the site of duodenal diverticula.
4	B	EASY	Milky spots are macrophage patches in the greater omentum, the basis of its role as the 'policeman of the abdomen'. Peyer's patches are ileal lymphoid tissue.
5	D	MEDIUM	The gastroduodenal artery is a posterior relation of the first part; about 65% of duodenal ulcers are on this posterior wall and can erode it.
6	B	MEDIUM	A posterior gastric ulcer can erode into the pancreas (back pain) and the splenic artery running along it in the stomach bed, causing severe haemorrhage.
7	A	MEDIUM	Gastric cancer spreads via lymphatics; cells reach the left supraclavicular (Virchow's) node through the thoracic duct. A palpable node is Troisier's sign.
8	E	MEDIUM	In the free margin: bile duct anterior-right, hepatic artery proper anterior-left, portal vein posterior and deepest. The gastric vessels run in the hepatogastric ligament.
9	C	MEDIUM	The stomach receives sympathetic fibres from T6–T9 (greater splanchnic nerves, celiac plexus); these segments also supply the upper abdominal wall.
10	D	MEDIUM	Anterior trunk = left vagus → anterior surface; posterior trunk = right vagus → mainly posterior surface. Cutting them reduces parietal-cell acid secretion.
11	A	MEDIUM	Jejunum: wider, thicker, redder, few arcades, long vasa recta, less fat near the wall (white windows). The ileum has multiple arcades, short vasa recta and fat reaching the wall.
12	E	MEDIUM	The minor duodenal papilla (if present) lies ~2 cm above the major papilla and receives the accessory pancreatic duct. Bile and main pancreatic ducts open at the major papilla (ampulla of Vater).
13	B	MEDIUM	The ligament of Treitz is a fibromuscular band from the right crus to the duodenojejunal flexure; it stops the flexure being dragged down and is the upper/lower GI landmark.
14	C	MEDIUM	The left lobe of the liver is an ANTERIOR relation. Stomach bed: spleen, left suprarenal, upper left kidney, splenic artery, pancreas, transverse mesocolon and colon.
15	E	HARD	SMA syndrome: the third part crosses L3 between the SMA (anterior) and aorta (posterior). Suspect it in young patients with rapid weight loss and post-prandial pain and vomiting.
16	A	HARD	The first inch is intraperitoneal; anterior perforation spills into the greater sac above the transverse colon, then down the right lateral paracolic gutter, mimicking appendicitis.
17	D	HARD	The third part runs horizontally across the L3 vertebral body and is most exposed; the second part lies in the paravertebral gutter and is the most protected.
18	B	HARD	Lower half of the duodenum: inferior pancreaticoduodenal artery (SMA) and lymph down to the superior mesenteric nodes. The upper half follows the celiac route.
19	C	HARD	The root crosses, from above down: third part of duodenum, aorta, IVC, right gonadal vessels, right ureter, right psoas major. All crossed structures (except the midline ones) are on the RIGHT.
20	D	HARD	The right gastro-omental artery (from the gastroduodenal) is ligated this way in antrectomy; the rich anastomoses among the gastric arteries keep the remnant perfused.